

Amelia Eckard

Charlotte, NC | ameliaeckard.com | github.com/ameliaeckard | linkedin.com/in/ameliaeckard

Education

University of North Carolina at Charlotte <i>M.S. in Artificial Intelligence / Early Entry Program</i>	Charlotte, NC <i>Expected May 2028</i>
University of North Carolina at Charlotte <i>B.S. in Computer Science / Concentration: AI, Robotics, & Gaming</i>	Charlotte, NC <i>Expected May 2027</i>

Experience

Undergraduate Researcher <i>UNC Charlotte, Dr. Todd Dobbs</i>	Charlotte, NC <i>August 2025 – May 2026</i>
- Developed an Apple Vision Pro indoor-navigation prototype for visually impaired users using Swift, ARKit, RealityKit, ObjectTrackingProvider, Core ML, HRTF spatial audio, and distance-based pitch cues. - Investigated object tracking, on-device recognition, and spatial-audio guidance as complementary modalities for accessible indoor navigation. Designed evaluation protocols for object identification, tracking reliability, navigation performance, and spatial-audio feedback workflows.	

Research Intern <i>Stevens Institute of Technology, Dr. Jina Huh-Yoo</i>	Remote <i>June 2026 – Present</i>
Engineered <i>rq-system</i>/LEAF, an AI-assisted literature-analysis pipeline using PubMed E-utilities, CrossRef, and LLM APIs to screen, classify, and extract structured evidence from 345 papers; processed 256 included, 80 excluded, and 9 review-flagged papers.	

Lead Instructional Assistant, ITSC 1213 <i>UNC Charlotte College of Computing and Informatics</i>	Charlotte, NC <i>August 2026 – Present</i>
Serve as lead instructional assistant for an asynchronous Python programming course, coordinating grading, rubrics, office hours, student support, and course logistics with the instructor.	

Projects

CNN Scene Classification <i>Python, PyTorch, torchvision, CUDA</i>	2026
- Trained and evaluated convolutional neural networks for 16-class scene recognition on 2,400 images. - Improved validation accuracy from 42.25% to 51.46% through image augmentation while comparing preprocessing choices, training behavior, and model performance.	
Emergency Fund Predictor <i>Python, scikit-learn, pandas, SHAP</i>	2025
Built an 11-feature machine-learning pipeline using 12,295 Federal Reserve SHED respondents enriched with BEA indicators to study emergency-fund vulnerability.	
Vision Lab: ASCII Reconstruction <i>JavaScript, HTML/CSS, Image Processing</i>	2026 – Present
- Developed an image-to-ASCII reconstruction experiment for a web-based computer-vision lab connected to ameliaeckard.com. - Designed reconstruction logic around luminance mapping, local contrast, detail preservation, and edge-aware character selection to retain recognizable image structure.	
Iris <i>Electron, Node.js, Discord.js, SQLite, sqlite-vec, Ollama, MariaDB</i>	2025 – Present
Developed a local-first personal-memory and community-automation system combining desktop tooling, persistent memory, local language models, and Discord operations.	
Resolve – LPL Financial University Hackathon <i>AI/ML, Workflow Intelligence, FinTech</i>	2026
Developing a proactive exception-intelligence layer that predicts likely blockers before financial-service requests visibly stall.	

Technical Skills

Languages	Swift, Python, C++, JavaScript
Spatial/AI	ARKit, RealityKit, Core ML, OpenCV, ObjectTrackingProvider, HRTF spatial audio
Tools	Git, CUDA, NumPy, PyTorch, HTML/CSS

Certifications & Involvement

UR2PhD Training Course, Computing Research Association	December 2025
Social and Behavioral Research, CITI Program	September 2025